

Lab #6: Controlling an Inverted Pendulum

Released: Mar 09, 2026

Due: Mar 11/12, 2026

This lab is a repeat of Lab #5 so that everyone can finish it.

Objectives

By the end of this lab, you should be able to

1. Learn how to linearize a nonlinear system and derive the state-space representation of the system.
2. Appreciate the importance of linearization in the process of control design and how the controller synthesized using the linearized model can be used to control the original nonlinear system.

1 Assigned Setup: Cart or Rotary?

Ask your TA which setup you are assigned to work on for the lab. The two setups are the cart-pole system (also called the single pendulum gantry) and the rotary pendulum system. Depending on the setup you are assigned to, you should read the background section for that setup to understand the physical system and the mathematical model of the system.

The goal for both setups is the same — to linearize the nonlinear equations of motion of the system to obtain the state-space representation of the system.

Note: There is no “pre-lab” work associated with this lab that needs to be done before going to the lab. You can perform all calculations in the lab. Estimate that you will need about 1 hour (at most) to perform the linearization and then around 30 minutes to collect the data and the last 30 minutes to finish your report.

2 Background: Cart Pendulum

The cart pendulum system, also called the cart-pole system is a system where a cart moves in one dimension to control the angle of the pendulum. In this lab, we call

this the linear single pendulum gantry (SPG) system. Note that this is not because it is a “linear system”, the linear here refers to the 1-D motion. A physical model of the cart-pole SPG system is shown in Figure 1.

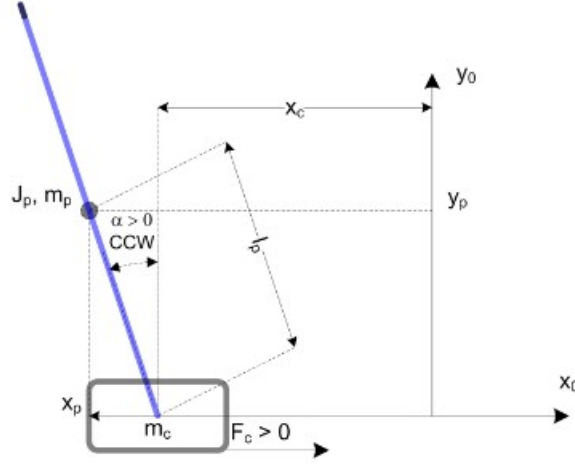


Figure 1: The cart-pole system (single pendulum gantry)

The pendulum pivot is on the IP02 cart, and is measured using the **Pendulum** encoder. The pendulum has mass, M_p , the centre of mass of the pendulum is at length, l_p , and the moment of inertia about the centre of mass is J_p . The pendulum angle, α , is zero when it is suspended perfectly vertically and increases positively when rotated counter-clockwise (CCW). The positive direction of linear displacement of the cart, x_c , is to the right when facing the cart. The position of the pendulum centre of gravity is denoted as the (x_p, y_p) coordinate.

2.1 The nonlinear model of the cart-pole system

The nonlinear equations of motion for the cart and for the pendulum, respectively, are:

$$(J_{eq} + M_p)\ddot{x}_c + M_p l_p \cos(\alpha)\ddot{\alpha} - M_p l_p \sin(\alpha)\dot{\alpha}^2 = F_c - B_{eq}\dot{x}_c \quad (1)$$

and

$$M_p l_p \cos(\alpha)\ddot{x}_c + (J_p + M_p l_p^2)\ddot{\alpha} + M_p l_p g \sin(\alpha) = -B_p \dot{\alpha} \quad (2)$$

where F_c is the linear force applied to the cart, M_p is the mass of pendulum, l_p is the length of center of mass of pendulum from pivot, J_{eq} is the effective mass of the

cart, J_p is the moment of inertia of the pendulum about the center of mass, g is the acceleration due to gravity, and B_{eq} is the equivalent viscous damping coefficient. This is our control input to the system.

Pre-Lab Quiz 2.1: Make sure you're ready for the lab!

What is the order of the cart-pole system? List all state variables.

- A. First-order
- B. Second-order
- C. Third-order
- D. Fourth-order

The equations of motion are nonlinear due to the presence of the trigonometric functions of α and the product of $\dot{\alpha}^2$ with $\sin(\alpha)$.

Your main goal in the lab is to linearize this model to obtain the state-space matrices A , B , C , and D for the system. That is, the goal is to obtain the linear system model of the form:

$$\dot{x} = Ax + Bu \quad (3)$$

and

$$y = Cx + Du \quad (4)$$

where x is the state, u is the control input, and A , B , C , and D are the state-space matrices. The state x is a 4×1 vector given by the cart position x_c , pendulum angle α , cart speed $\dot{x}_c$, and pendulum angular velocity $\dot{\alpha}$ and can be written as the vector:

$$x = \begin{bmatrix} x_c \\ \alpha \\ \dot{x}_c \\ \dot{\alpha} \end{bmatrix} \quad (5)$$

The output y of the system is partial, and consists of the measurements of cart position and the pendulum angle:

$$y = \begin{bmatrix} x_c \\ \alpha \end{bmatrix} \quad (6)$$

Pre-Lab Quiz 2.2: Make sure you're ready for the lab!

What are the dimensions of the state-space matrices A , B , C , and D for the cart-pole system?

Based on the output description above, the matrices C and D are:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (7)$$

and

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (8)$$

Pre-Lab Quiz 2.3: Make sure you're ready for the lab!

Without finding out the matrices A and B , prove that given the definition of the state vector x and the output y , the matrices C and D must be as shown above.

NOTE: After this point, you can jump to the linearization steps if you have been assigned the cart-pole system. The parameters of your cart-pole system are all stored in the setup file on the lab computer. You can run the file and get the values of all constants from MATLAB.

3 Background: Rotary Pendulum

The rotary pendulum system uses a rotary arm (servo) to control the angle of the pendulum. The physical model of the rotary pendulum system is shown in Figure 2.

The two nonlinear equations of motion for the rotary arm and for the pendulum, respectively, are:

$$\begin{aligned} & \left(m_p L_r^2 + \frac{1}{4} m_p L_p^2 - \frac{1}{4} m_p L_p^2 \cos(\alpha)^2 + J_r \right) \ddot{\theta} - \left(\frac{1}{2} m_p L_p L_r \cos(\alpha) \right) \ddot{\alpha} \\ & + \left(\frac{1}{2} m_p L_p^2 \sin(\alpha) \cos(\alpha) \right) \dot{\theta} \dot{\alpha} + \left(\frac{1}{2} m_p L_p L_r \sin(\alpha) \right) \dot{\alpha}^2 = \tau - B_r \dot{\theta} \end{aligned} \quad (9)$$

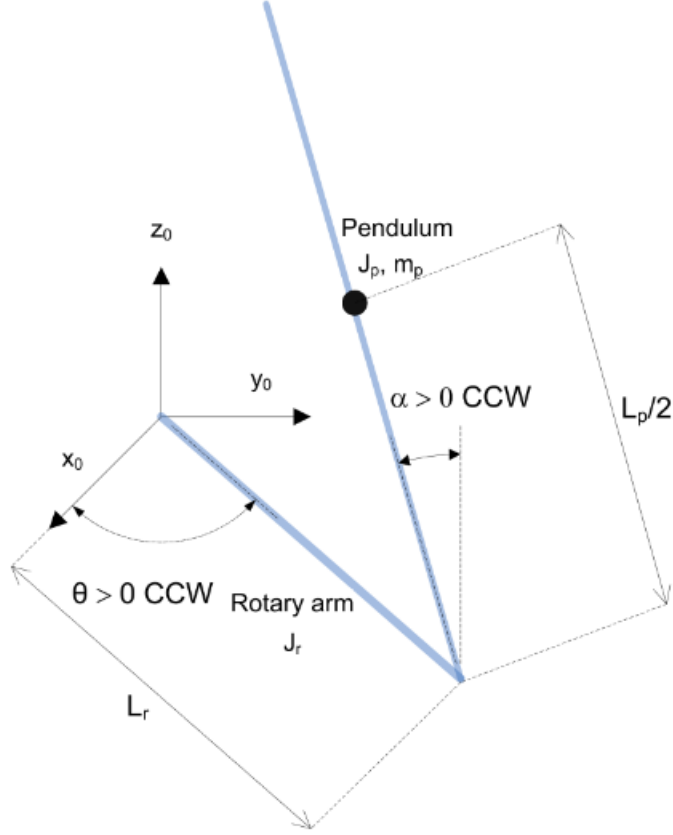


Figure 2: The rotary pendulum system

$$\begin{aligned}
 -\frac{1}{2}m_p L_p L_r \cos(\alpha) \ddot{\theta} + \left(J_p + \frac{1}{4}m_p L_p^2 \right) \ddot{\alpha} - \frac{1}{4}m_p L_p^2 \cos(\alpha) \sin(\alpha) \dot{\theta}^2 \\
 - \frac{1}{2}m_p L_p g \sin(\alpha) = -B_p \dot{\alpha}. \quad (10)
 \end{aligned}$$

where θ is the angle of the rotary arm, α is the angle of the pendulum, m_p is the mass of the pendulum, L_r is the length of the rotary arm, L_p is the length of the pendulum, J_r is the moment of inertia of the rotary arm about its pivot, J_p is the moment of inertia of the pendulum about its center of mass, g is the acceleration due to gravity, B_r is the viscous damping coefficient of the rotary arm, B_p is the viscous damping coefficient of the pendulum, and τ is the torque applied to the rotary arm.

This is our control input to the system.

$$\tau = \frac{\eta_g K_g \eta_m k_t (V_m - K_g k_m \dot{\theta})}{R_m}. \quad (2.4)$$

The equations of motion are nonlinear due to the presence of the trigonometric functions of α and the product of $\dot{\theta}^2$ with $\cos(\alpha)\sin(\alpha)$ and the product of $\dot{\theta}\dot{\alpha}$ with $\sin(\alpha)\cos(\alpha)$.

Pre-Lab Quiz 3.1: Make sure you're ready for the lab!

What is the order of the rotary pendulum system? List all state variables.

- A. First-order
- B. Second-order
- C. Third-order
- D. Fourth-order

Your main goal in the lab is to linearize this model to obtain the state-space matrices A , B , C , and D for the system. That is, the goal is to obtain the linear system model of the form:

$$\dot{x} = Ax + Bu \quad (11)$$

and

$$y = Cx + Du \quad (12)$$

where x is the state, u is the control input, and A , B , C , and D are the state-space matrices. The state x is a 4×1 vector given by

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} := \begin{bmatrix} \theta \\ \alpha \\ \dot{\theta} \\ \dot{\alpha} \end{bmatrix}, \quad u := \tau. \quad (13)$$

So, we implicitly have

$$\dot{x}_1 = x_3, \quad \dot{x}_2 = x_4. \quad (14)$$

The output y of the system is partial, and consists of the measurements of servo angle and the pendulum angle:

$$y = \begin{bmatrix} \theta \\ \alpha \end{bmatrix} \quad (15)$$

Pre-Lab Quiz 3.2: Make sure you're ready for the lab!

What are the dimensions of the state-space matrices A , B , C , and D for the cart-pole system?

Based on the output description above, the matrices C and D are:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (16)$$

and

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (17)$$

Pre-Lab Quiz 3.3: Make sure you're ready for the lab!

Without finding out the matrices A and B , prove that given the definition of the state vector x and the output y , the matrices C and D must be as shown above.

4 Experimental Steps: Linearization

To linearize the nonlinear equations of motion given in equations (1) and (2) for cart-pole and in equations (9) and (10) for rotary pendulum, follow the steps as shown in the class:

Step 1: Rewrite in standard form

Rewrite the equations in the standard form discussed in class $\dot{x} = f(x, u)$. Recall that the state vector for the cart-pole is:

$$x = \begin{bmatrix} x_c \\ \alpha \\ \dot{x}_c \\ \dot{\alpha} \end{bmatrix} \quad \text{OR} \quad x = \begin{bmatrix} \theta \\ \alpha \\ \dot{\theta} \\ \dot{\alpha} \end{bmatrix}$$

and the control input is $u = F_c$ (for cart-pole) OR $u = \tau$ (for rotary pendulum). Using equations (1) and (2) for cart-pole OR using equations (9) and (10) for rotary pendulum, solve for the second derivatives $\ddot{x}_c$ or $\ddot{\theta}$, and $\ddot{\alpha}$ in terms of the state variables and control input. This will give you expressions of the form:

$$\begin{aligned} \dot{x}_1 &= x_3 \\ \dot{x}_2 &= x_4 \\ \dot{x}_3 &= f_3(x_1, x_2, x_3, x_4, u) \\ \dot{x}_4 &= f_4(x_1, x_2, x_3, x_4, u) \end{aligned}$$

Step 2: Identify the equilibrium point

Determine the equilibrium point (x_e, u_e) about which to linearize. For the inverted pendulum, we want to balance the pendulum in the upright position with the cart at rest. What are the values of x_c , α , $\dot{x}_c$, $\dot{\alpha}$, and F_c at this equilibrium for the cart-pole? OR For the rotary pendulum, what are the values of θ , α , $\dot{\theta}$, $\dot{\alpha}$, and τ at the equilibrium?

Hint: Notice in Figures 1 and 2 that the pendulum angle is measured from the upright position. So, the equilibrium point for the pendulum angle is $\alpha = 0$ (not π).

Step 3: Compute the Jacobian to get the A matrix

The A matrix is the Jacobian of $f(x, u)$ with respect to x , evaluated at the equilibrium point:

$$A = \left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix}_{(x_e, u_e)}$$

Compute each partial derivative carefully. You are allowed to use computer code to perform the differentiation / compute the Jacobian. After finding the derivatives, don't forget to substitute the equilibrium values to evaluate the Jacobian at the equilibrium point.

Step 4: The input matrix B

The B matrix is the Jacobian of $f(x, u)$ with respect to u , evaluated at (x_e, u_e) :

$$B = \left. \frac{\partial f}{\partial u} \right|_{(x_e, u_e)} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \\ \frac{\partial f_3}{\partial u} \\ \frac{\partial f_4}{\partial u} \end{bmatrix}_{(x_e, u_e)}$$

Compute each partial derivative with respect to the control input $u = F_c$ and evaluate at the equilibrium point.

Step 5: Verify your results

Check that your A and B matrices have the correct dimensions (4×4 and 4×1 , respectively) and verify that:

- The first two rows of A correspond to the trivial relationships $\dot{x}_c = \dot{x}_c$ and $\dot{\alpha} = \dot{\alpha}$ OR $\dot{\theta} = \dot{\theta}$ and $\dot{\alpha} = \dot{\alpha}$, which means that the first two rows of A should have a 1 in the column corresponding to $\dot{x}_c$ or $\dot{\theta}$ and $\dot{\alpha}$, respectively, and zeros elsewhere.
- The first two elements of B are zero (since F_c does not directly affect $\dot{x}_c$ or $\dot{\alpha}$). For rotary pendulum, since τ does not directly affect $\dot{\theta}$ or $\dot{\alpha}$, the first two elements of B should be zero.
- The linearized system captures the unstable nature of the inverted pendulum (hint: check the eigenvalues of A : at least one should have a positive real part).

Pre-Lab Grade and Experimental Steps
Due: Mar 11/12, 2026 Name: _____ Grade: _____

1 Hands-on Experiment: Group Work

On the lab computer, make sure that you change the workspace directory to the folder on the Desktop labeled “EE 122” and under that “Lab #5”. Open the setup file for your assigned equipment and the associated Simulink model: `ip02_quick_start` for those working on cart-pole and for `q_rotpen_bal` for those working on the rotary pendulum.

1. Set the values of the state-space matrices that you obtained in the simulation part of the lab in `SIP_ABCD_eqn_student.m` (for those on the cart-pole setup) OR `ROTPEN_ABCD_eqns_student.m` (for those on the rotary pendulum setup). AFTER setting the matrix values, run the setup script.
- 2 Follow the steps as previous labs on the Simulink model: “Build for Monitoring”, then “Connect”, then “Start”.
- 3 To test whether the controller works, slowly bring the pendulum to an upright position and keep it there until you observe the controller take over. If the controller is working, you should see the pendulum stabilize at the upright position and the cart move to the desired position / rotary servo move to stabilize the pendulum. However, if the controller feels strange, don’t force it or leave it alone. Bring the pendulum down to zero and try again. If it still doesn’t work, stop the simulation and check your state-space matrices and gain values.
4. Open the cart position scope / rotary angle scope, the motor input voltage scope, and the pendulum angle scope to view the graphical data in real-time.
5. Identify which color trace corresponds to what variable in the scope.
6. Once you have observed the response, click “Stop” to stop the simulation.
7. Add a small perturbation to the system matrix A and observe how the response changes. Try two perturbations and describe how the response changes compared to the original response for each. You are allowed to choose the kinds of

perturbations you apply to the matrix A but discuss your strategy of what you experimented, what you predicted would happen, and what actually happened.

8. Export the data from the MATLAB workspace variables to your personal computer to analyze the response further and to understand the impact of the modeling error perturbations you applied to the system in the previous step. Note that the data will be saved under the **Lab #5** folder in the **Lab_5_data** folder.

2 Action Items

In your lab report, provide the following:

1. Show the derivation / approach of linearization to find the matrices A , B , C , and D for the state-space representation of the system. You are allowed to use MATLAB code for linearization or do it by hand. In either case, you must show your work.
2. Write down the original A matrix and the two perturbed A matrices (ones with modeling error that you introduced).
3. For each of the three A matrices, show the plots of the response of the system.
4. Reflect on the differences in the response of the system for each of the three A matrices. Describe how the perturbations you applied to the system affected the response of the system and whether it met the performance requirements.

Discuss your results in detail here.